

MATH 1610 / HONORS CALCULUS

Honors Calculus 1610

Lecture 5: Area, Step Functions, and the Integral

September 18, 2026

Where we are

Course outline

- ◆ Chunk 1: precalculus
- ◆ Chunk 2: calculus (1 variable), plus a bit of vectors
- ◆ Chunk 3: multivariable

Last time

- ◆ Finished the real numbers (Dedekind cuts).
- ◆ Cartesian products, ordered pairs, functions.

Before the exam

- ◆ (Definite) integrals.
- ◆ Limits and continuity.
- ◆ Intermediate value theorem, etc.

Story time: the complex and hypercomplex numbers

First: a square root of -1

At some point it became useful to have a square root of -1 :

$$i^2 = -1.$$

The standard representation of the resulting system is $\mathbb{R}$ together with i , with

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Axiomatizing $\mathbb{C}$

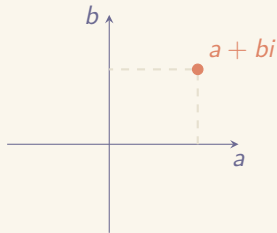
A complex number is a pair of real numbers (a, b) , with

$$(a, b) + (c, d) = (a + c, b + d),$$

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc),$$

and the identification

$$(a, b) \longleftrightarrow a + bi.$$



Two dimensions, then three?

The complex numbers were a revolution in two-dimensional geometry in the 1700s. What about three dimensions?

Hamilton found that the answer lives in four dimensions: the quaternions $\mathbb{H}$, with i, j, k and

$$\begin{aligned} i^2 = j^2 = k^2 = -1, \\ ij = k = -ji, \quad jk = i = -kj, \quad ki = j = -ik. \end{aligned}$$

Multiplication is not commutative.

Quaternion arithmetic, in full

$$(a + bi + cj + dk) + (e + fi + gj + hk) = (a + e) + (b + f)i + (c + g)j + (d + h)k,$$

$$\begin{aligned}(a + bi + cj + dk)(e + fi + gj + hk) \\&= (ae - bf - cg - dh) \\&\quad + (af + be + ch - dg) i \\&\quad + (ag - bh + ce + df) j \\&\quad + (ah + bg - cf + de) k.\end{aligned}$$

Quaternions and rotations

Write a vector in three-space as

$$v = ai + bj + ck,$$

and let $q = d + ei + fj + gk$ be a quaternion. Then

$$q v q^{-1} = \text{“}v \text{ rotated by } q\text{”}.$$

This is where the i, j, k notation for vectors comes from.

And then it stops: the octonions

A few years later came the **octonions** $\mathbb{O}$, in **eight** dimensions: seven imaginary units $e_1, \dots, e_7$, each with $e_n^2 = -1$, multiplied according to a fixed rule.

Multiplication there is **not even associative**.

They are related to rotations in 7 and 8 dimensions — 8 in particular.

$$\mathbb{R} \subset \mathbb{C} \subset \mathbb{H} \subset \mathbb{O}$$

Back to (pre)calculus

Warm up: which of these are functions?

1) $\{(x, y) \in \mathbb{Z} \times \mathbb{R} \mid y = x^2\}$

2) $\{(x, y) \in \mathbb{R} \times \mathbb{Z} \mid y = x^2\}$

3) $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = x^2\}$

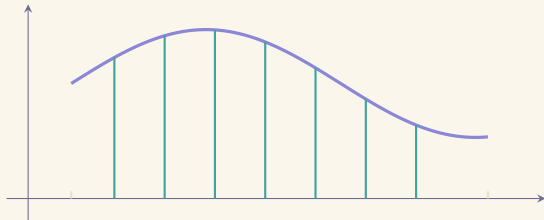
4) $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y^2 = x\}$

5) $\{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x - y \text{ is even}\}$

Recall: $f: X \rightarrow Y$ is a subset $f \subseteq X \times Y$ such that for every x there is **exactly one** y with $(x, y) \in f$.

Area (§1.6)

The integral we are after is the **area under a graph**.



But what *is* area? Before defining it we should be warned that “size” can behave very badly.

A first warning

$$\mathbb{Z} = \{\text{even integers}\} \cup \{\text{odd integers}\},$$

yet

$$\#\{\text{even integers}\} = \#\mathbb{Z}.$$

A set can be split into pieces each of which is “as big as” the whole.

Banach–Tarski

Let $S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$.

One can find subsets

$$A_1, A_2, B_1, B_2, B_3 \subseteq S,$$

all mutually disjoint, whose union is S , and **rigid rotations and translations** $A_i \rightsquigarrow A'_i$, $B_j \rightsquigarrow B'_j$, such that

$$A'_1 \cup A'_2 = S, \quad B'_1 \cup B'_2 \cup B'_3 = S.$$

So the pieces cannot all have a well-defined area.

The concept: measurability

We are not going to say, once and for all, which sets have an area. What we write down instead is an **axiomatic framework** for what it means to *have* a concept of measurability: a special collection of subsets, whose members we call **measurable**, together with an **area function** on them.

$$\mathcal{M} = \text{some set of subsets of } \mathbb{R}^2, \quad a: \mathcal{M} \longrightarrow \mathbb{R},$$

so $a(S) \in \mathbb{R}$ for S measurable.

The axioms below do **not** by themselves pick out a single such concept.

The axioms

- ◆ If S, T are measurable, so are $S \cup T$, $S \cap T$, $S \setminus T$, and

$$a(S \cup T) = a(S) + a(T) - a(S \cap T).$$

- ◆ $a(S) \geq 0$ for every measurable S .
- ◆ $a(\emptyset) = 0$.
- ◆ a is invariant under translation.
- ◆ The unit square has area 1.
- ◆ Exhaustion.

etc.

From these one *deduces* that a rectangle with sides a and b has area ab — it is not an axiom.

Exhaustion

Axiom

Let $T \subseteq \mathbb{R}^2$ and set

$$A = \{a(S) \mid S \subseteq T, S \text{ measurable}\}, \quad B = \{a(S) \mid T \subseteq S, S \text{ measurable}\}.$$

If $\sup(A) = \inf(B) = L$, then T is measurable and $a(T) = L$.

This is the moral of Dedekind cuts again: we pin a real number down by what lies above it and what lies below it, and if the two agree, we have our number.

Step functions

Definition

$f: [a, b] \rightarrow \mathbb{R}$ is a **step function** if there are

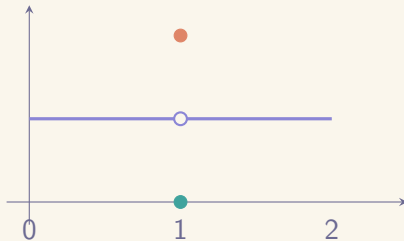
$$x_0, x_1, \dots, x_n \in [a, b], \quad x_0 = a, \quad x_n = b, \quad x_i < x_{i+1},$$

with $f(x) = c_i$ constant for $x \in (x_{i-1}, x_i)$.

Note the **open** intervals: the values at the subdivision points are not mentioned, and will not matter.

The values at the jumps do not matter

Take $f(x) = 1$ for $x \in [0, 2] \setminus \{1\}$, and let $f(1)$ be whatever you like — say 2, or 0.



A single point contributes no area, and the answer is 2 either way.

The integral of a step function

Definition

If f is a step function as above,

$$\int_a^b f(x) dx = \sum_{i=1}^n c_i (x_i - x_{i-1}).$$

When $f \geq 0$ this is exactly the area of the rectangles under the graph, so it agrees with the area function of the previous slides.

Properties of the integral of a step function

- ◆ **Linearity:** $\int_a^b (f + g) = \int_a^b f + \int_a^b g$, and $\int_a^b \lambda f = \lambda \int_a^b f$.
- ◆ **Monotonicity:** if $f(x) \leq g(x)$ for all x , then $\int_a^b f \leq \int_a^b g$.
- ◆ **Additivity over intervals:** $\int_a^b f + \int_b^c f = \int_a^c f$.
- ◆ $\int_a^a f = 0$, so it is consistent to define $\int_b^a f = -\int_a^b f$ and have the integral defined for any ordering of the endpoints.

Integrability

Let $f: [a, b] \rightarrow \mathbb{R}$ be a function, and set

$$L = \sup \left\{ \int_a^b s(x) dx \mid s \text{ a step function, } s \leq f \right\},$$

$$U = \inf \left\{ \int_a^b t(x) dx \mid t \text{ a step function, } f \leq t \right\},$$

where $s \leq f$ means $s(x) \leq f(x)$ for all x .

Definition

If $U = L$ we say f is **integrable**, and define $\int_a^b f(x) dx = L = U$.

Aside

If $f(x) \geq 0$ always, then $\int_a^b f(x) dx$, if it exists, is the area under the curve — in the sense of the exhaustion axiom, with the step functions below and above playing the roles of A and B .

These all hold for arbitrary integrable functions

Suppose f and g are integrable, with

$$s_1 \leq f \leq t_1, \quad s_2 \leq g \leq t_2,$$

step functions, and write $L_1 = \int_a^b f$, $L_2 = \int_a^b g$.

Then $s_1 + s_2 \leq f + g \leq t_1 + t_2$, and

$$\int_a^b s_1 \leq L_1, \quad \int_a^b s_2 \leq L_2.$$

Additivity, concluded

Given any $\varepsilon > 0$, choose step functions s_1, s_2 with

$$\int_a^b s_1 > L_1 - \frac{\varepsilon}{2}, \quad \int_a^b s_2 > L_2 - \frac{\varepsilon}{2}.$$

Then $s_1 + s_2$ is a step function below $f + g$, and

$$\int_a^b (s_1 + s_2) = \int_a^b s_1 + \int_a^b s_2 > L_1 + L_2 - \varepsilon.$$

Since ε was arbitrary, the lower integral of $f + g$ is $L_1 + L_2$; the same argument from above finishes it. $\square$

Integrating a power

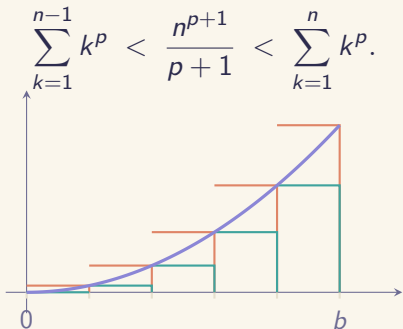
Theorem (1.15)

If p is a positive integer and $b > 0$, then

$$\int_0^b x^p dx = \frac{b^{p+1}}{p+1}.$$

The inequality behind it

Cut $[0, b]$ into n equal pieces and compare x^p with the step functions below and above it. Everything comes down to



Proof of the inequality, by induction

Write $m = n - 2$, and let $P(m)$ be the statement

$$\sum_{k=1}^{m+1} k^p < \frac{(m+2)^{p+1}}{p+1}.$$

Base case $P(0)$:

$$1 < \frac{2^{p+1}}{p+1}, \quad \text{that is,} \quad p+1 < 2^{p+1}.$$

A side induction

Let $Q(p)$ be the statement $p + 1 < 2^{p+1}$.

$Q(0)$: $1 < 2$. ✓

$Q(p + 1)$:

$$(p + 1) + 1 < 2^{p+1} \cdot 2 = 2^{p+1} + 2^{p+1},$$

which holds since $p + 1 < 2^{p+1}$ and $1 < 2^{p+1}$. $\square$

The inductive step

Assume $P(m)$. Then

$$\sum_{k=1}^{m+2} k^p = \sum_{k=1}^{m+1} k^p + (m+2)^p < \frac{(m+2)^{p+1}}{p+1} + (m+2)^p.$$

On the other hand, expanding the binomial,

$$\begin{aligned} \frac{((m+2)+1)^{p+1}}{p+1} &= \frac{(m+2)^{p+1}}{p+1} + \frac{(p+1)(m+2)^p}{p+1} + \dots \\ &= \frac{(m+2)^{p+1}}{p+1} + (m+2)^p + \dots \end{aligned}$$

and every remaining term is positive. This gives $P(m+1)$. $\square$

Next time

Limits and continuity.